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École Polytechnique de Louvain

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Project - Aerial Image Segmentation

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Group 01

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Part A - Supervised Segmentation Task

1.1 Network Architectures

Two architectures were selected for comparative evaluation: improved version of U-Net and DeepLab model.

1.1.1 U-Net Architecture

A symmetric encoder–decoder with skip connections.

- **Encoder (4 levels):** each level applies two Conv+BN+ReLU blocks.
  - Channels: 3→64, 64→128, 128→256, 256→512 (with pooling at each stage)
- **Bottleneck:** 1024 channels with dropout to reduce overfitting.
- **Decoder:** up-convolution (×2), skip fusion, optional attention, two Conv+BN blocks.
- **Attention:** enhances feature merging and improves boundary accuracy.

1.1.2 DeepLab Architecture

Designed to improve multi-scale reasoning through:

- **Dilated convolutions (atrous):** enlarge the receptive field without increasing parameters.
- **Spatial Pyramid Pooling (ASPP):** captures multi-scale context by partitioning the feature map into grids, max-pooling each region, and concatenating all outputs into a fixed-size representation, enabling scale-invariant semantic aggregation.

Encoder: three strided convolutional stages before ASPP.

1.2 Network Training

- **Loss:** cross-entropy for pixel-wise multiclass segmentation, minimizing the divergence between predicted probabilities and ground-truth labels, and naturally handling five-class semantic masks.
- **Optimization:** Adam (adaptive per-parameter learning rate, fast convergence, robust to initialization), with additional experiments using SGD as a baseline to compare optimization dynamics and generalization stability.
- **Hyper-parameters:** best configuration identified via grid search: LR = 0.001, batch size = 16. Training was conducted for up to 100 epochs with early stopping (patience = 10) to prevent overfitting while allowing sufficient convergence time.

1.3 Evaluation

Several quantitative and qualitative metrics were implemented to assess the segmentation performance:

- **Quantitative Metrics:**
  - **Accuracy** Measures the proportion of correctly classified pixels.
  - **Weighted Precision** Evaluates the model's ability to avoid false positives.
  - **Weighted Recall** Measures the ability to detect all true positive pixels.
  - **Weighted F1-Score** Harmonic mean of precision and recall.
  - **Intersection over Union (IoU)** Computed per class.
  - **ROC-AUC** Discriminative capacity of the model by class.
- **Qualitative Metrics:**

- **Image:** input RGB image used for visual inspection of segmentation quality.
- **Mask:** ground truth semantic labels serving as reference for pixel-wise evaluation.
- **Overlay:** superposition of predicted labels on the original image to visually assess boundary accuracy and spatial consistency.
- **Prediction:** output segmentation map generated by the model, illustrating class assignment and misclassification regions.

1.4 Results

1.4.1 Comparison of Architectures

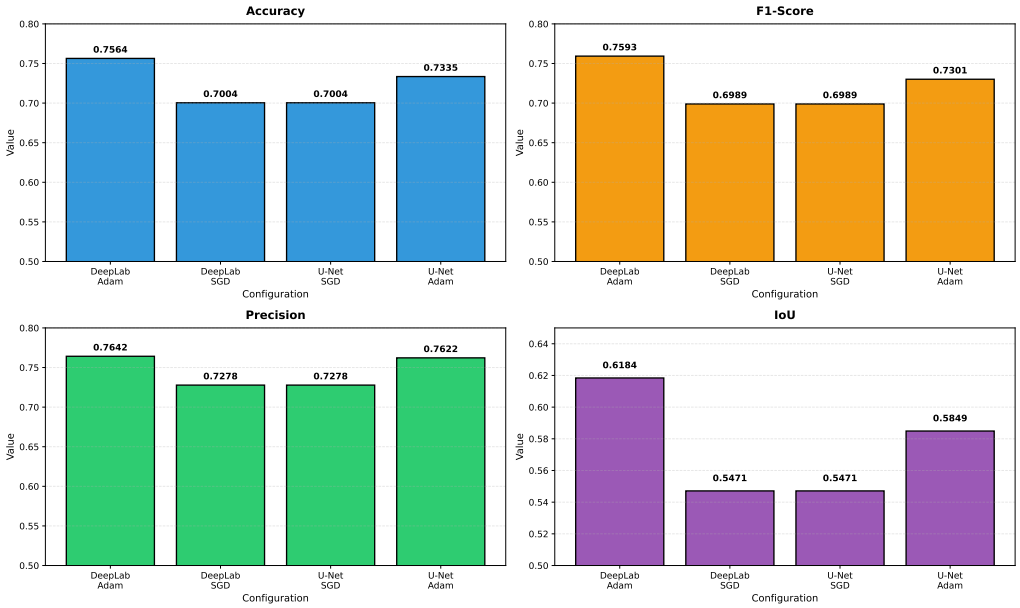


Figure 1.1: Performance comparison of DeepLab and U-Net with Adam and SGD using Accuracy, Precision, F1-Score and IoU.

With SGD, both architectures achieve similar performance, meaning fixed learning-rate updates do not strongly benefit either model. Using Adam clearly improves all metrics for both networks, with DeepLab achieving the best overall results. Adam's adaptive updates enhance convergence and stability, making DeepLab + Adam the most effective configuration.

1.4.2 DeepLab with Adam Optimizer

The exact same graphs are available for U-Net with Adam Optimizer in the Appendice.

Figure 1.2: Class-wise performance metrics for the DeepLab architecture trained with the Adam optimizer, showing Precision, Recall, F1-Score and IoU for each semantic class.

Class-wise metrics highlight strong performance heterogeneity:

- High-performing classes (0 = Others, 2 = Buildings, 4 = Green Spaces):
  - **Class 0** achieves excellent precision and recall with an F1-Score of 82.3%, indicating reliable detection and minimal confusion with other classes.
  - **Classes 2 and 4** also show robust segmentation with balanced precision/recall around 70%, suggesting that buildings and green spaces present clear spatial patterns. Some confusion with Class 0 remains, reflecting semantic overlap or similar textures at certain scales.
- Challenging classes (1 = Water and 3 = Farmlands):
  - **Water regions (Class 1)** present the lowest precision (18.4%) and IoU (12.1%), showing frequent confusion with surrounding areas. This class is strongly underrepresented in the dataset, making it harder for the model to learn distinctive features and leading to poor discriminability.
  - **Farmlands (Class 3)** exhibits the opposite pattern: high recall (60.3%) but low precision (45.3%), meaning most true instances are detected but with many false positives. Class imbalance and texture similarity with other land-use regions further reinforce these errors, although the class is slightly better represented than Class 1.

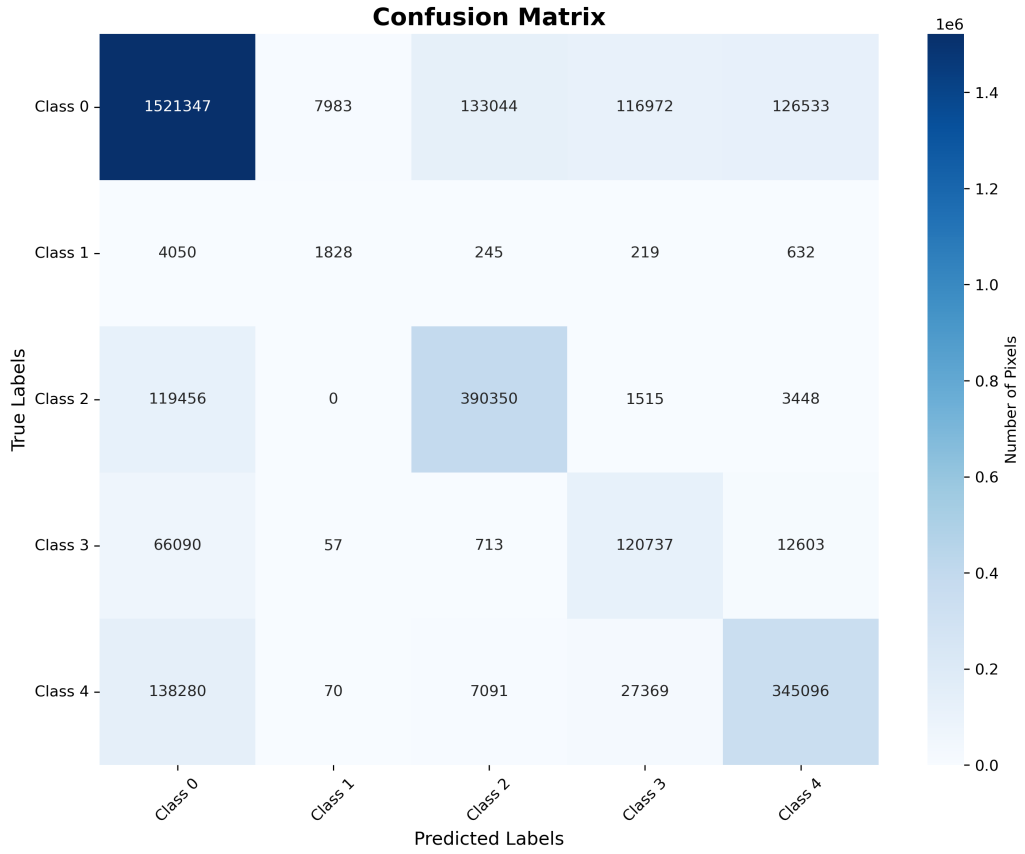


Figure 1.3: Class-wise confusion matrix for DeepLab with Adam, showing the number of correctly and incorrectly predicted pixels per class and highlighting inter-class confusion patterns.

Confusion matrices confirm the same trends:

- Largest diagonal values are observed for Classes 0, 2 and 4, confirming reliable segmentation and class separability.
- Class 1 and Class 3 show strong off-diagonal activation, indicating systematic overlap with visually similar regions, especially vegetation or terrain textures.

This confirms that dataset imbalance and spatial ambiguity strongly impact class discrimination.

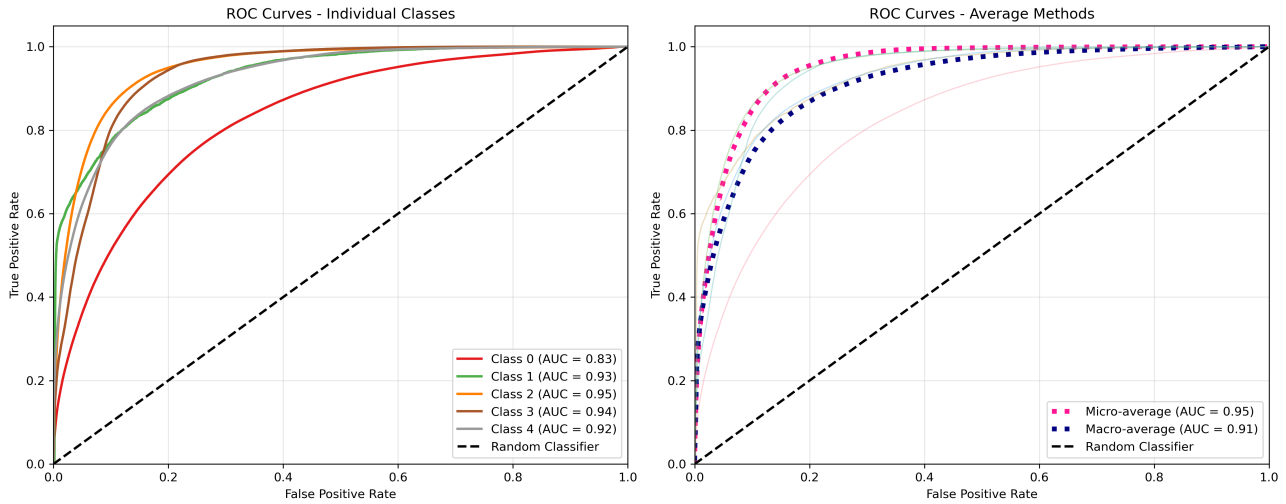
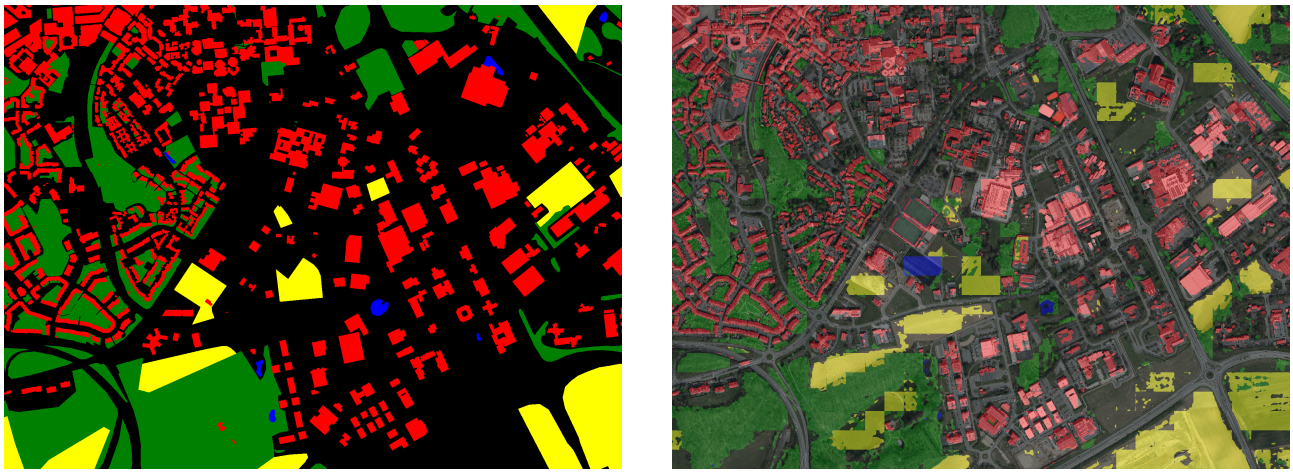


Figure 1.4: ROC curves for DeepLab with Adam. The figure includes class-wise ROC performance and micro/macro-averaged curves, illustrating the overall discriminative capability of the segmentation model.

ROC curves show strong discriminative capability overall, with micro and macro AUC above 90%, indicating that most classes can be separated reliably at the probability level. However, Class 0 displays the lowest class-wise AUC despite being the most represented. This is explained by its high intra-class variability: it groups heterogeneous objects and textures, making its decision boundary less sharp than for more semantically coherent classes. Minority or visually ambiguous classes also show degraded AUC, reflecting either class imbalance or semantic overlap.



(a) Ground truth mask

(b) Overlay: prediction vs input image

Figure 1.5: Qualitative comparison between ground truth mask and model prediction using overlay visualization for the DeepLab architecture with Adam.

Overlay visualizations indicate that global scene structures are well preserved: building blocks, green areas and urban layouts are accurately shaped and located. Errors mainly occur:

- around thin or transitional boundaries,
- in low-resolution or shadowed regions,
- and for rare classes with diffuse spatial features.

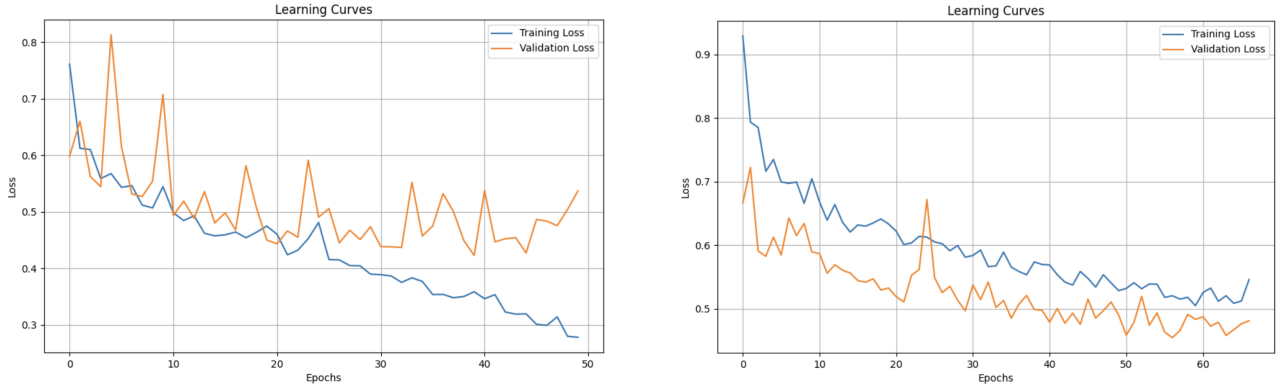
Nevertheless, the prediction remains coherent at global scale, and qualitative inspection confirms the numerical conclusions: DeepLab with Adam generalizes well to dominant classes, while misclassification is mainly localized to rare (Water region by example) or texture-ambiguous regions.

1.5 Impact of Data Augmentation

(a) Ground truth mask

(b) Overlay visualization with prediction

Figure 1.6: Qualitative comparison illustrating the effect of data augmentation on segmentation consistency for the DeepLab model.



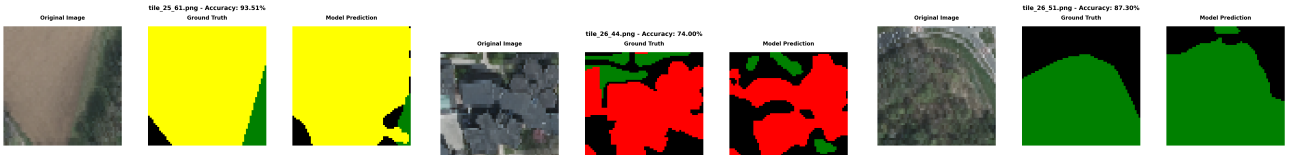
(a) Training without data augmentation

(b) Training with data augmentation

Figure 1.7: Learning curves with and without data augmentation for the DeepLab model, showing its effect on convergence stability and overfitting.

Data augmentation introduces geometric and photometric perturbations (flips, rotations, brightness/contrast and blur variations) during training. Figure 1.7 shows that augmentation provides smoother and more stable validation loss evolution, indicating reduced overfitting and improved robustness. However, this regularization slightly lowers peak performance metrics, since the model learns from more diverse but noisier samples. Overall, augmentation improves generalization to unseen data at the cost of a small decrease in maximal segmentation accuracy, which is a favorable trade-off in real-world aerial imagery applications.

1.6 Examples of Successful and Challenging Segmentations

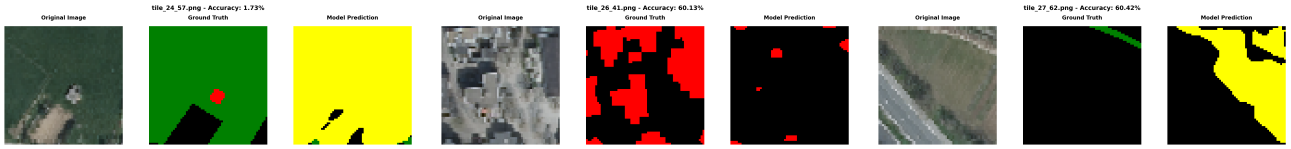


(a) Correct segmentation of homogeneous background

(b) Accurate building extraction

(c) Well-separated vegetation and non-vegetation areas

Figure 1.8: Examples of successful segmentation predictions from DeepLab with Adam.

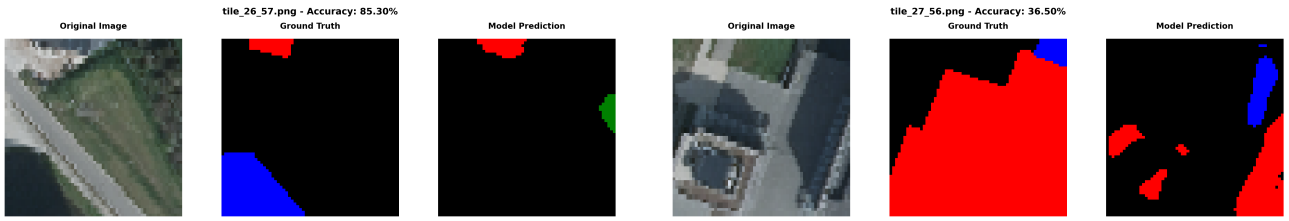


(a) Confusion between green spaces and farmlands

(b) Confusion between buildings and others

(c) Boundary errors in vegetation-covered areas

Figure 1.9: Challenging examples where visually similar classes reduce segmentation accuracy.



(a) Water region missed due to poor class representation

(b) Correctly segmented water body (partially)

Figure 1.10: Water detection examples highlighting class imbalance and visual ambiguity.

Good examples confirm that DeepLab can correctly segment major land-use patterns such as buildings and large vegetation areas, especially when structures are homogeneous or well defined. Challenging cases mainly arise from visually similar textures, particularly between green spaces and farmlands, leading to class switching without fundamentally damaging global scene structure. Water regions are the most difficult to detect: they are heavily underrepresented in the dataset and often appear in low-resolution or shadowed areas, which reduces discriminability and explains frequent false negatives. Overall, segmentation is globally coherent, with errors mostly localized to minority or ambiguous regions.

Part B - Self-Supervised Learning

2.1 Proxy Tasks

Three self-supervised proxy tasks were implemented to learn semantic features without manual labels:

- **Denoising:** Gaussian noise ($\sigma = 0.1$) is added to clean images. The encoder-decoder learns to reconstruct the original, forcing it to capture edges, textures and spatial structures. Loss used: MSE.
- **Colorization:** RGB images are converted to grayscale. The network predicts the original colors, requiring semantic understanding (water=blue, vegetation=green). Loss used: MSE.
- **Context Prediction:** 30% of the image is masked using 8×8 patches. The network reconstructs masked regions, learning spatial context and object continuity. Loss used: MSE.

2.2 Clustering

K-Means clustering was selected over DBSCAN for several reasons: (1) the dataset has five known semantic classes, making K-Means naturally aligned with the task; (2) K-Means is simpler and more stable than DBSCAN, which requires extensive hyperparameter tuning (ϵ, min_samples) and is sensitive to density variations; (3) K-Means allows easy exploration of different cluster counts (k=5, k=10) to investigate hidden subcategories. The configuration used is: StandardScaler normalization, $n_{init} = 20$, $max_{iter} = 500$, random_state=42.

2.3 Architectural Adaptations

The Part A encoder-decoder architecture was reused with minimal modifications:

- **Encoder:** Unchanged 4-level CNN (64→128→256→512 channels) with bottleneck (1024 channels). Extracts features at layer 4: $(B, 512, 8, 8)$ for 64×64 tiles.
- **Decoder:** Unchanged U-Net style with skip connections. Output layer adapted per task: 3 channels (denoising, colorization, context prediction).
- **Input modification:**
  - Denoising: $x_{noisy} = x_{clean} + \mathcal{N}(0, 0.1^2)$
  - Colorization: 1-channel grayscale input → 3-channel RGB output
  - Context Prediction: Image with masked patches (set to 0)
- **Padding:** Reflect padding to avoid border artifacts during clustering.

Note: All experiments were also reproduced using the DeepLab architecture, and are provided in the Appendix. The outputs can sometimes appear more visually noisy, but produce more coherent clusters in specific configurations than U-Net.

2.4 Proxy Task Results

2.4.1 Denoising

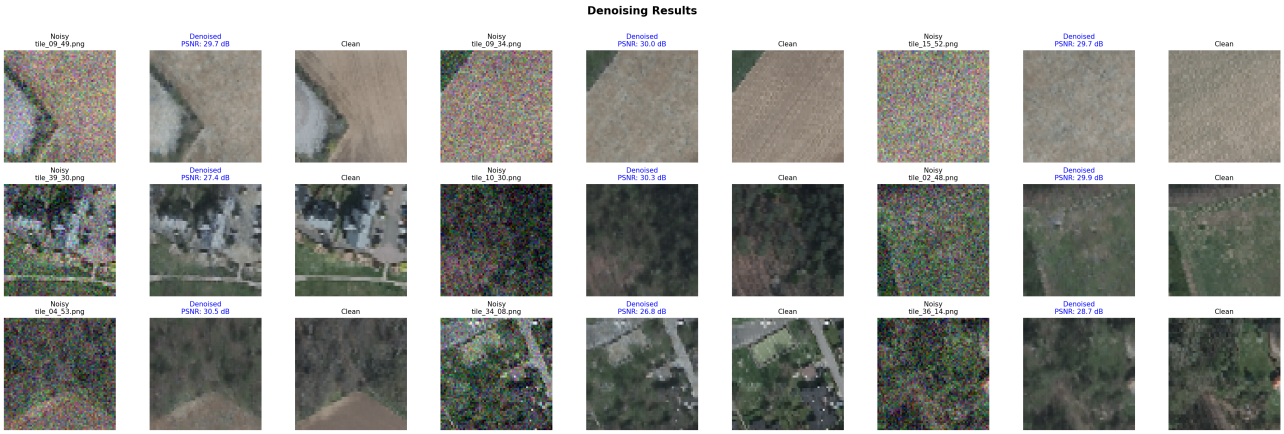


Figure 2.1: Denoising results: (left) noisy input, (center) prediction, (right) clean target.

2.4.2 Colorization

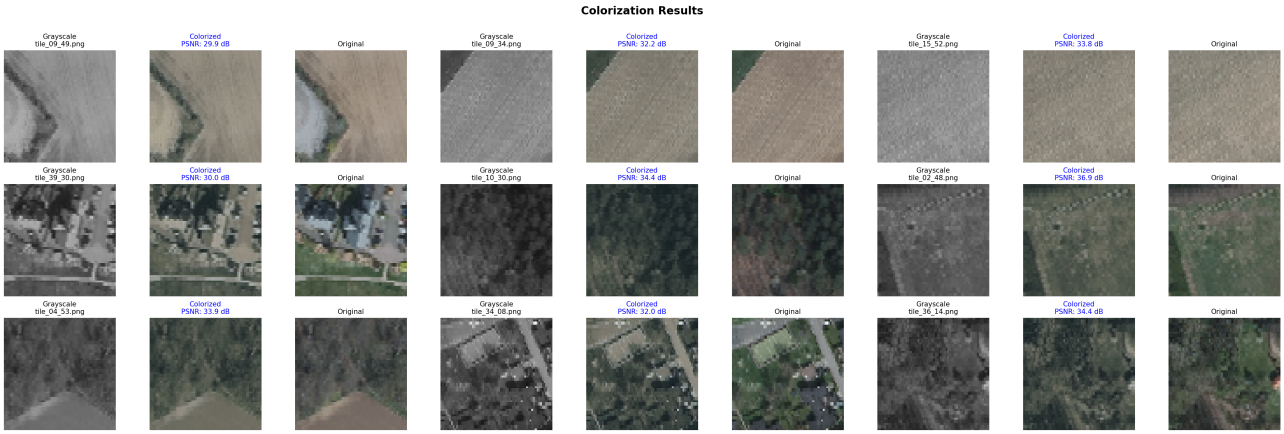


Figure 2.2: Colorization results: (left) grayscale input, (center) predicted colors, (right) original RGB.

2.4.3 Context Prediction

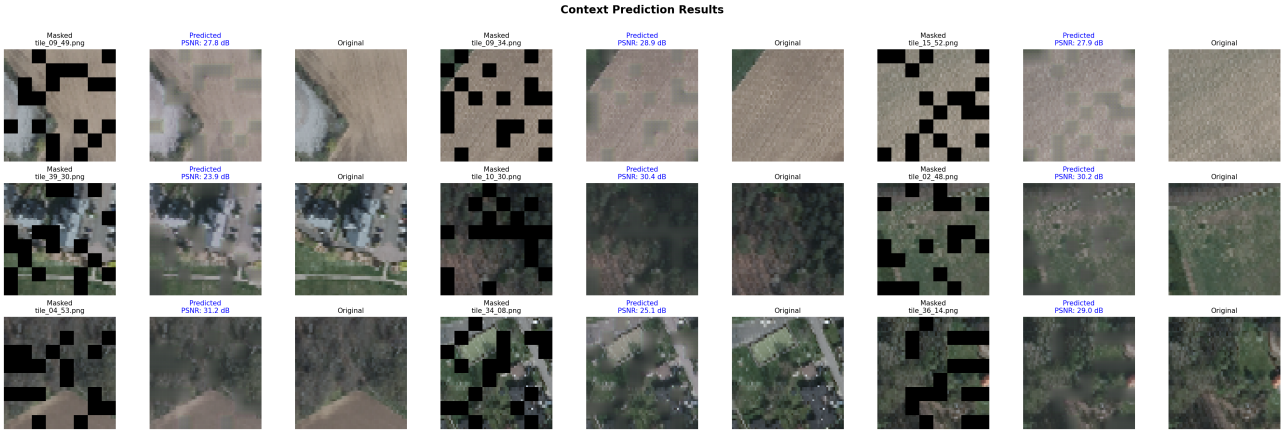


Figure 2.3: Context prediction: (left) masked input, (center) reconstruction, (right) original.

Interpretation: Across the three proxy tasks, qualitative results show distinct behaviors.

Denoising performs the best: the reconstructed outputs are close to the clean targets, with only minor residual noise. This indicates that the encoder learns meaningful low- and mid-level patterns that will likely transfer well to clustering.

Colorization is reasonably successful, although reconstructed images appear slightly desaturated and tend to bias toward dominant hues (e.g., whites turning yellow). This suggests that the model captures global structure and semantics but does not fully restore fine chromatic details.

Context prediction is clearly the most challenging: masked regions are filled with blurred textures instead of sharp reconstructions, meaning the network is unable to recover precise spatial information. As a consequence, we expect this proxy to produce weaker encoder features for downstream clustering.

2.5 Pseudo-Segmentation Results

2.5.1 Denoising (k=5)

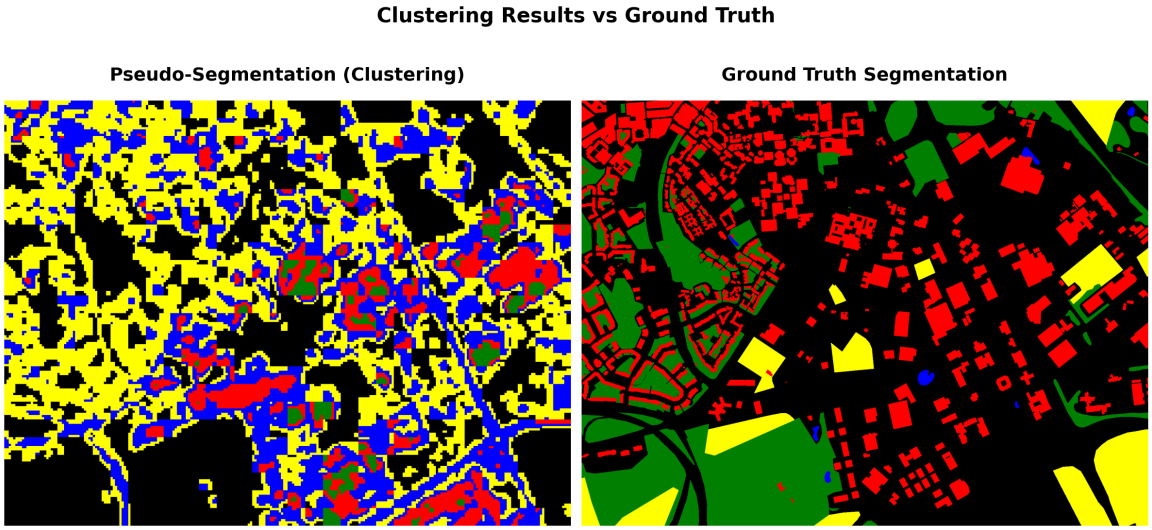


Figure 2.4: Pseudo-segmentation from denoising features (k=5): (left) clustering result, (right) ground truth.

2.5.2 Colorization (k=5)

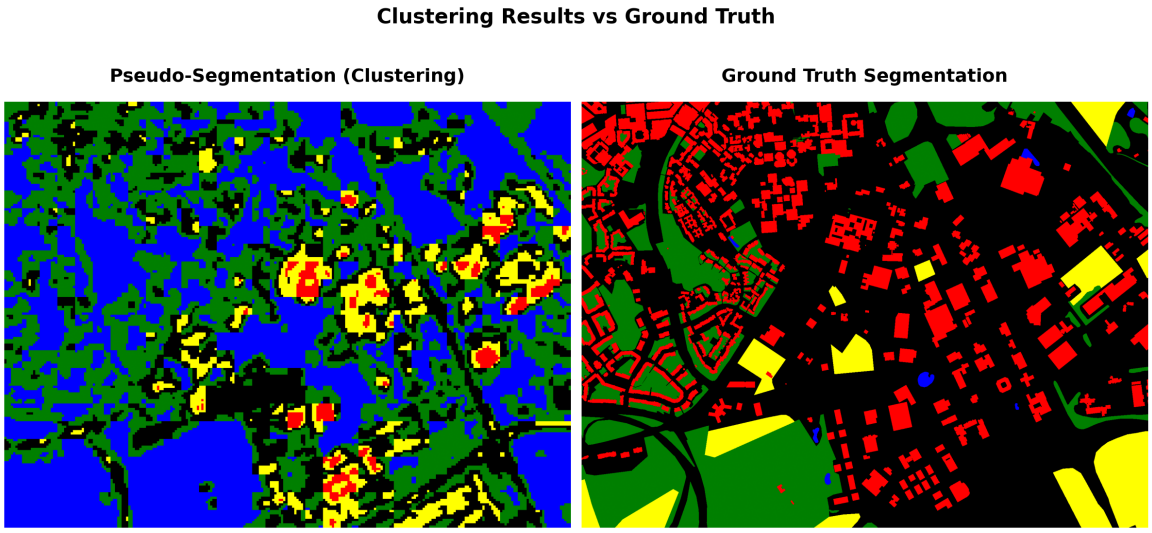


Figure 2.5: Pseudo-segmentation from colorization features (k=5): (left) clustering result, (right) ground truth.

2.5.3 Context Prediction (k=5)

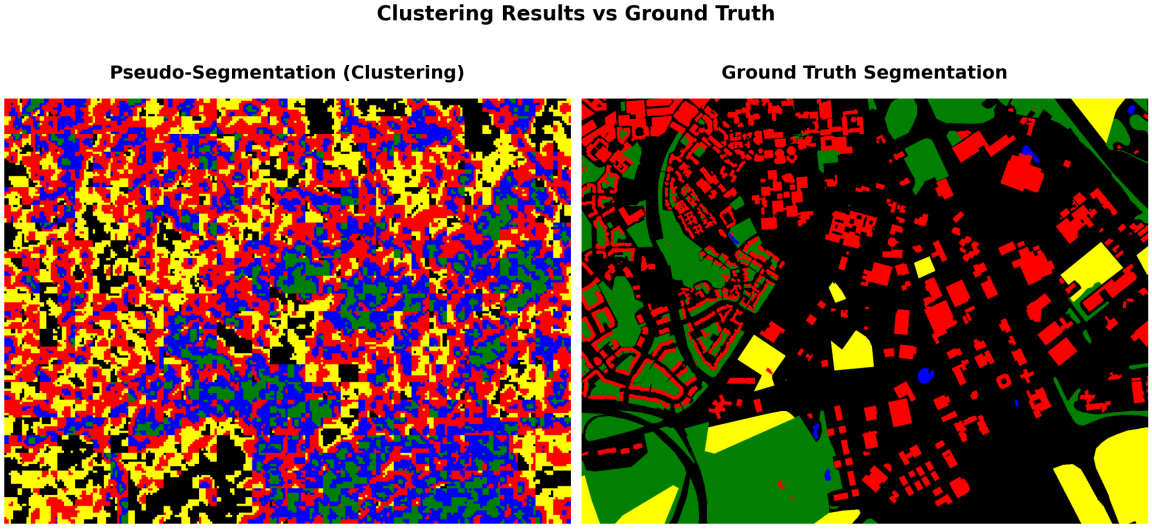


Figure 2.6: Pseudo-segmentation from context prediction features (k=5): (left) clustering result, (right) ground truth.

2.5.4 Denoising (k=10)

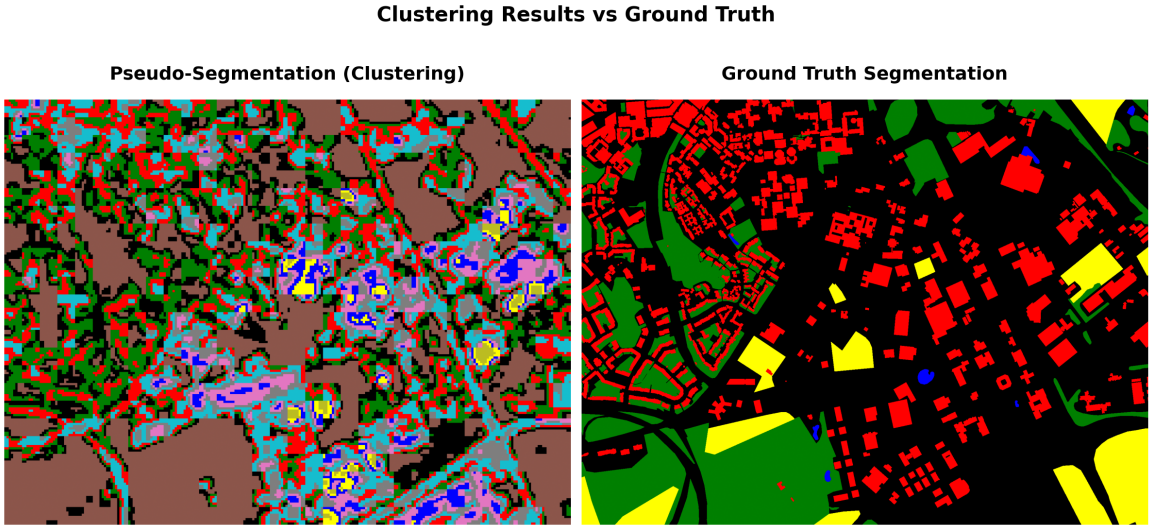


Figure 2.7: Pseudo-segmentation from denoising features (k=10): (left) clustering result, (right) ground truth.

2.5.5 Colorization (k=10)

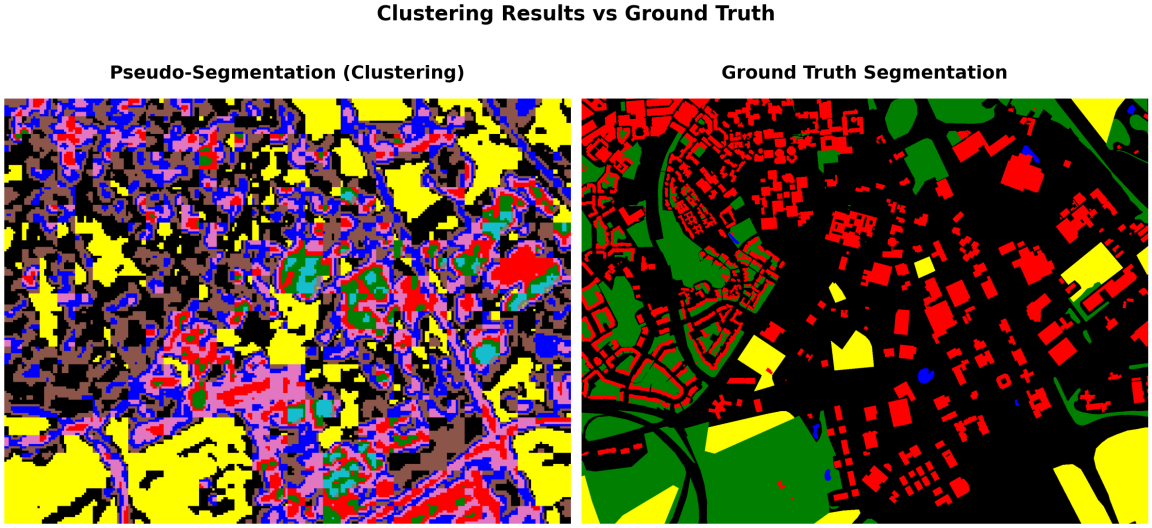


Figure 2.8: Pseudo-segmentation from colorization features (k=10): (left) clustering result, (right) ground truth.

2.5.6 Context Prediction (k=10)

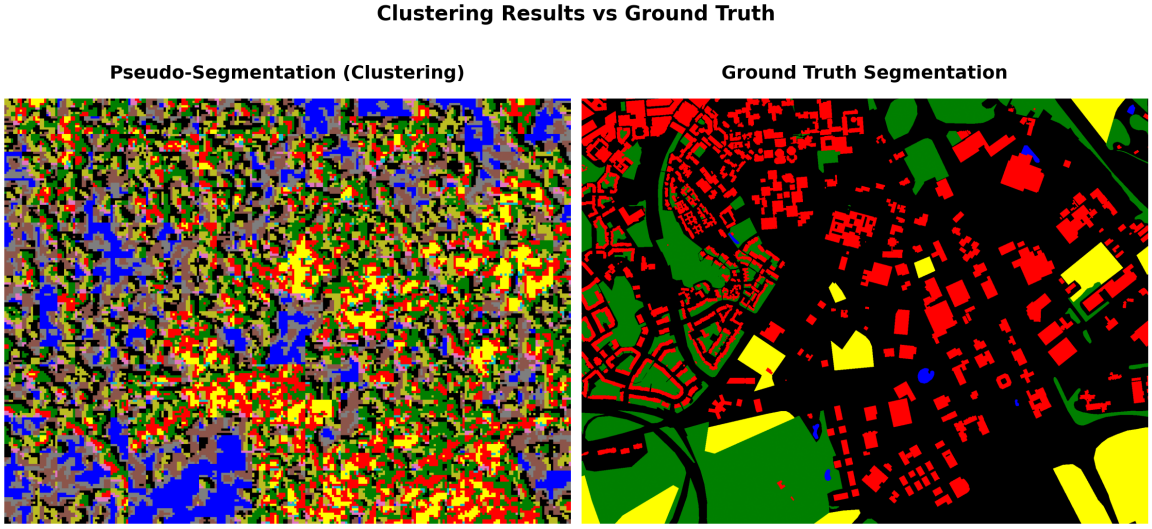


Figure 2.9: Pseudo-segmentation from context prediction features (k=10): (left) clustering result, (right) ground truth.

Interpretation (k=5): With $k = 5$, the clustering attempts to recover a segmentation structure similar to the ground truth. Since clustering is based on proxy-task features rather than labels, cluster colors do not correspond to semantic land-use classes.
Context prediction produces highly noisy and unstable pseudo-segmentations, confirming that this pretext task failed to learn meaningful spatial representations.
Denoising and colorization are more consistent: large vegetation regions and some linear structures (roads, rivers) are partially recognizable, and both methods yield visually similar pseudo-segmentation maps. However, object boundaries remain noisy and urban regions are split into multiple clusters, meaning that the extracted features capture broad spatial patterns but not fine semantic detail.

Interpretation (k=10): Increasing to $k = 10$ allows the models to capture additional internal variability. Context prediction remains unusable: clustering is extremely noisy and lacks spatial coherence.
Denoising and colorization again produce comparable results, with improved structure: roads and vegetation subtypes become more distinct, and urban areas split into multiple visual groups. These clusters do not correspond directly to semantic classes but reveal that richer latent structure is being extracted.
Overall, qualitative evaluation alone does not clearly separate denoising from colorization; quantitative metrics (e.g. ARI, NMI, IoU) would be required to determine which proxy task provides the most useful latent features.

Appendix

3.1 Part A - Improved U-Net Performance Graphs

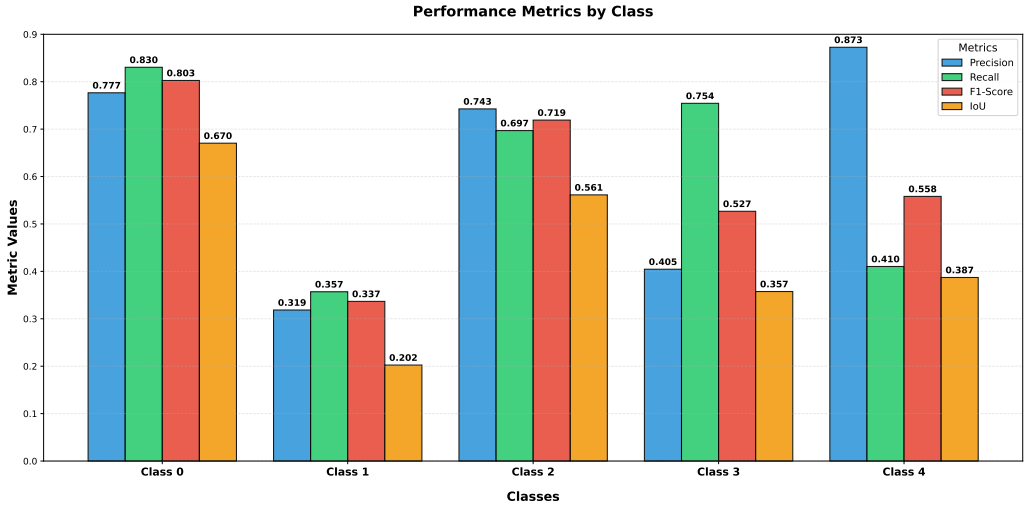


Figure 3.1: Class-wise evaluation metrics for the improved U-Net trained with Adam, reporting Precision, Recall, F1-Score and IoU per class.

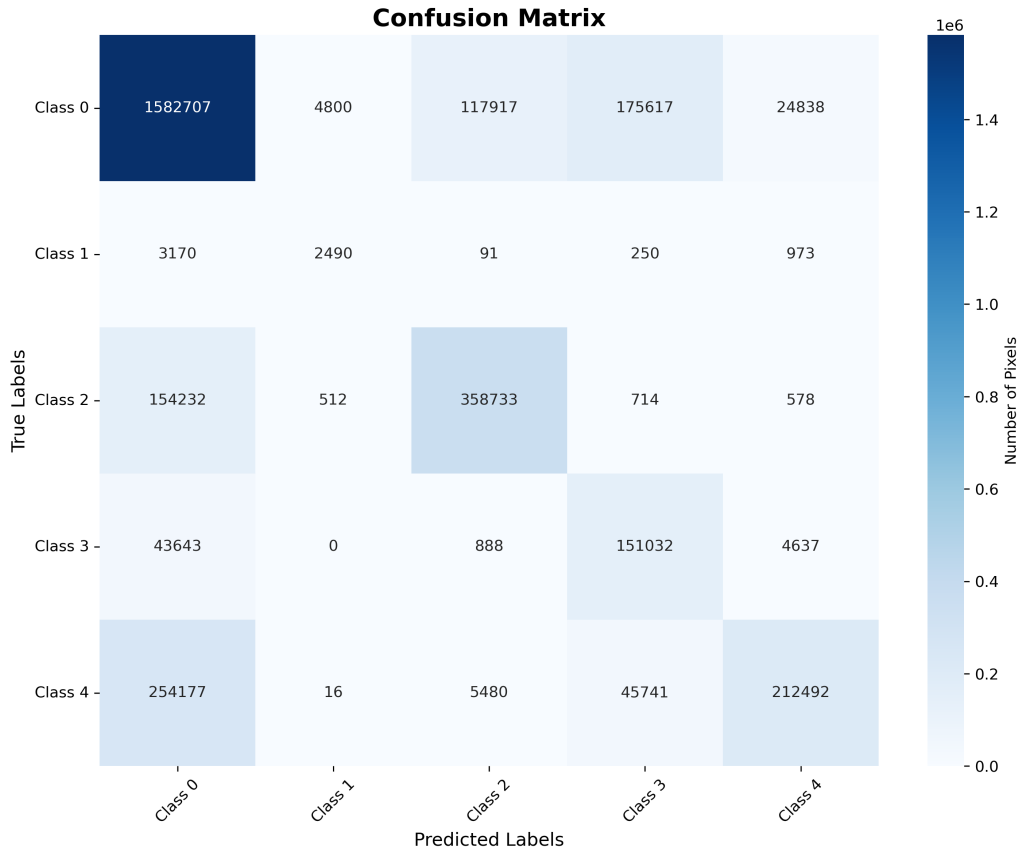


Figure 3.2: Class-wise confusion matrix for the improved U-Net trained with Adam, showing correctly and incorrectly predicted pixels per class and highlighting inter-class confusion.

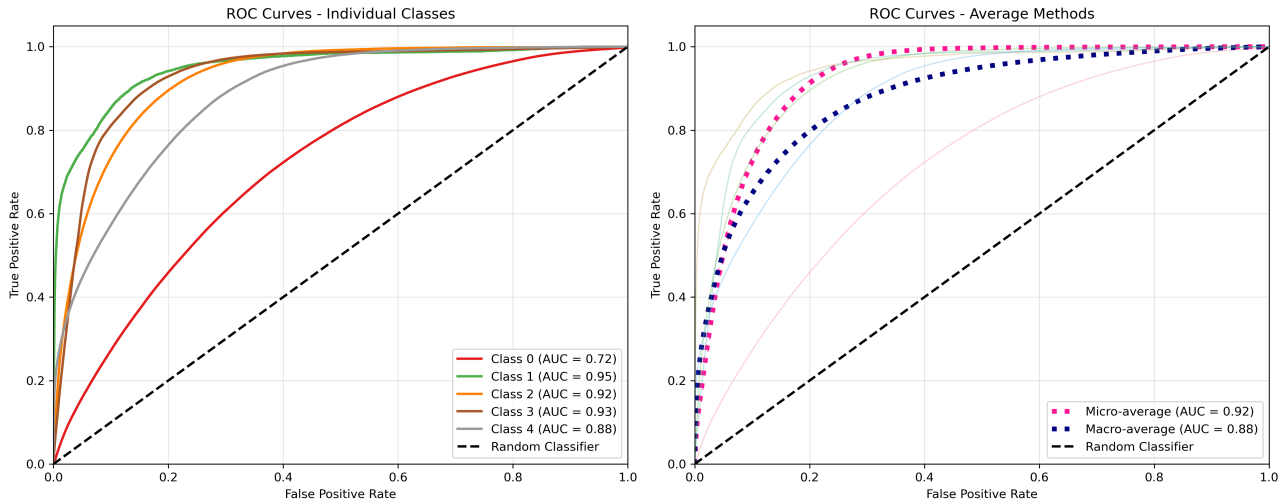
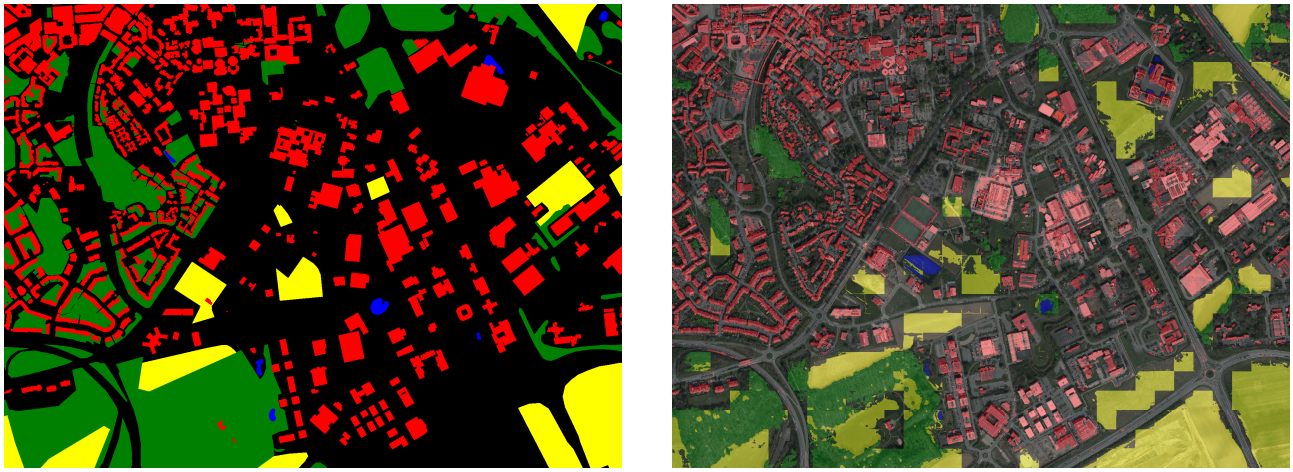


Figure 3.3: ROC curves for improved U-Net trained with Adam. The figure includes class-wise ROC performance and micro/macro-averaged curves, illustrating the overall discriminative capability of the segmentation model.



(a) Ground truth mask

(b) Overlay: prediction vs input image

Figure 3.4: Qualitative comparison between ground truth mask and overlay visualization for the improved U-Net trained with Adam.

3.2 Part B - DeepLab Performance Graphs

3.2.1 Proxy Task Results

3.2.1.1 Denoising

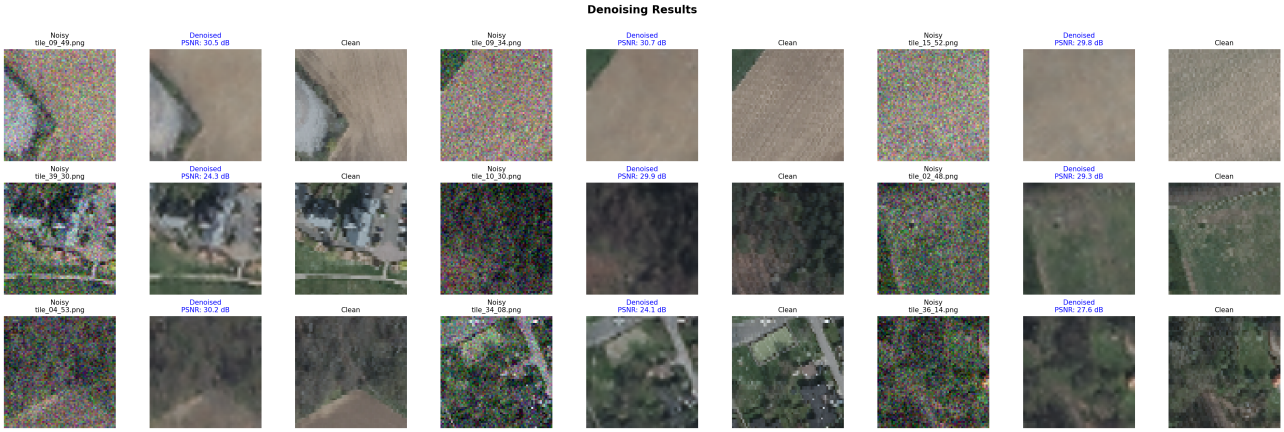


Figure 3.5: Denoising results: (left) noisy input, (center) prediction, (right) clean target.

3.2.1.2 Colorization

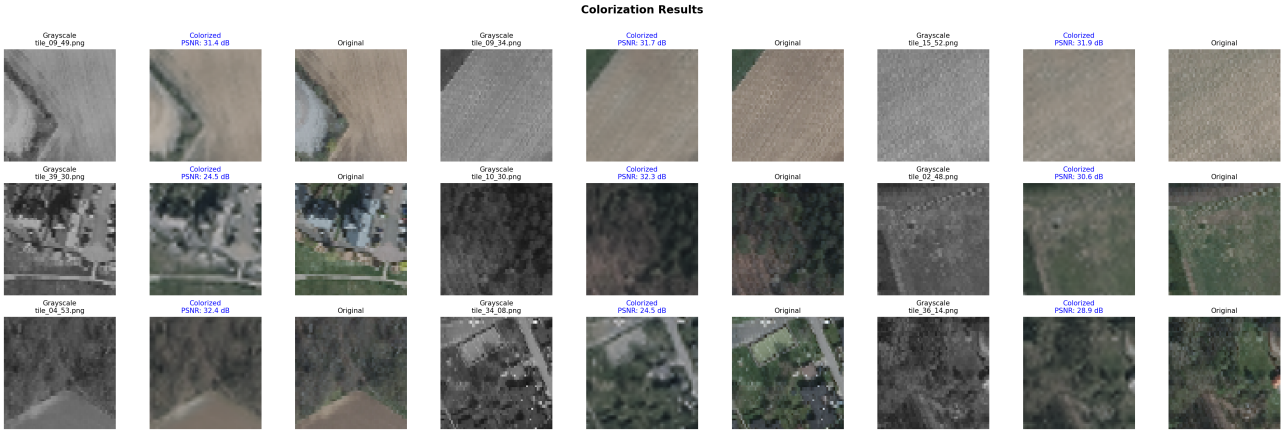


Figure 3.6: Colorization results: (left) grayscale input, (center) predicted colors, (right) original RGB.

3.2.1.3 Context Prediction

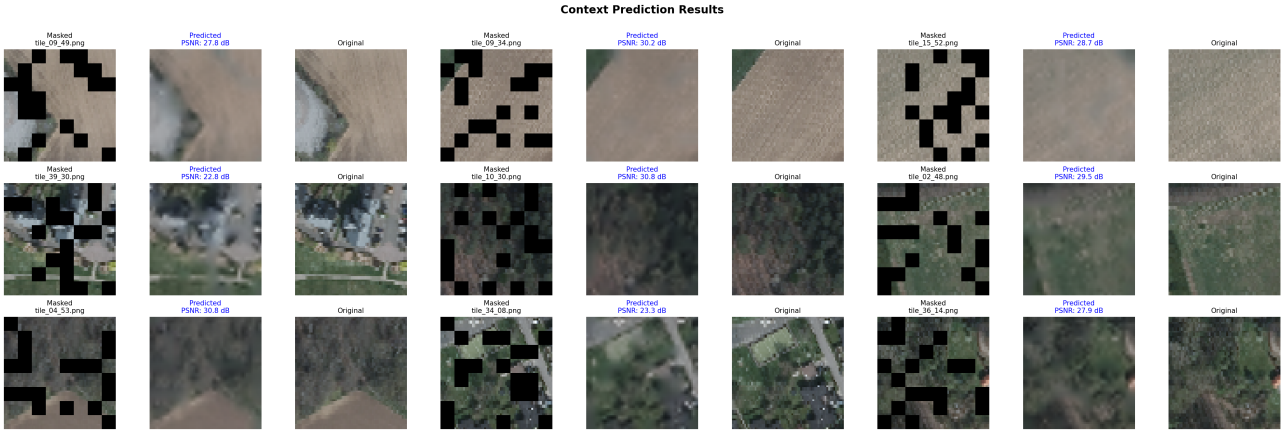


Figure 3.7: Context prediction: (left) masked input, (center) reconstruction, (right) original.

3.2.2 Pseudo-Segmentation Results

3.2.1.1 Denoising (k=5)

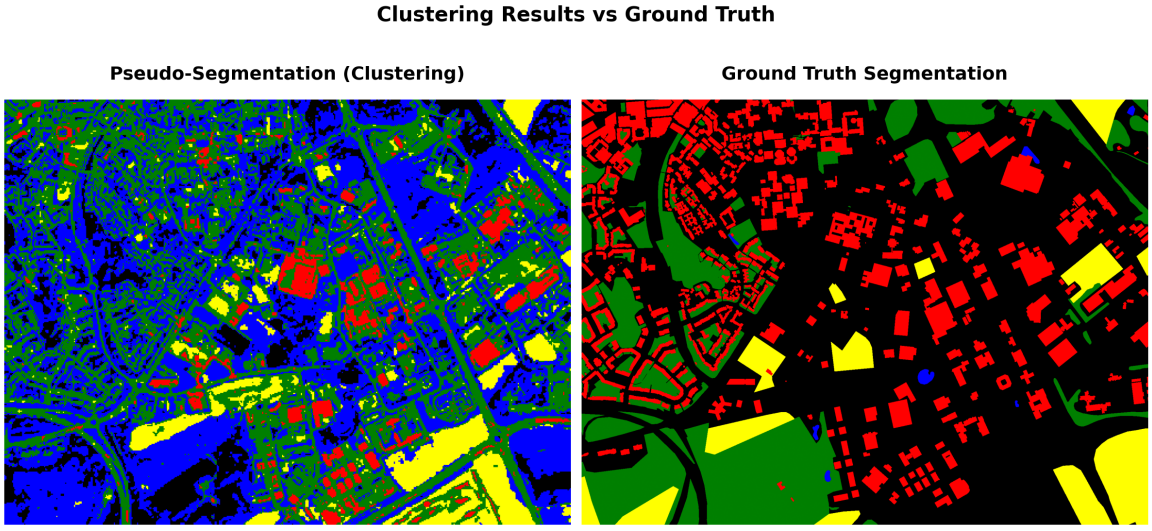


Figure 3.8: Pseudo-segmentation from denoising features (k=5): (left) clustering result, (right) ground truth.

3.2.2.2 Colorization (k=5)

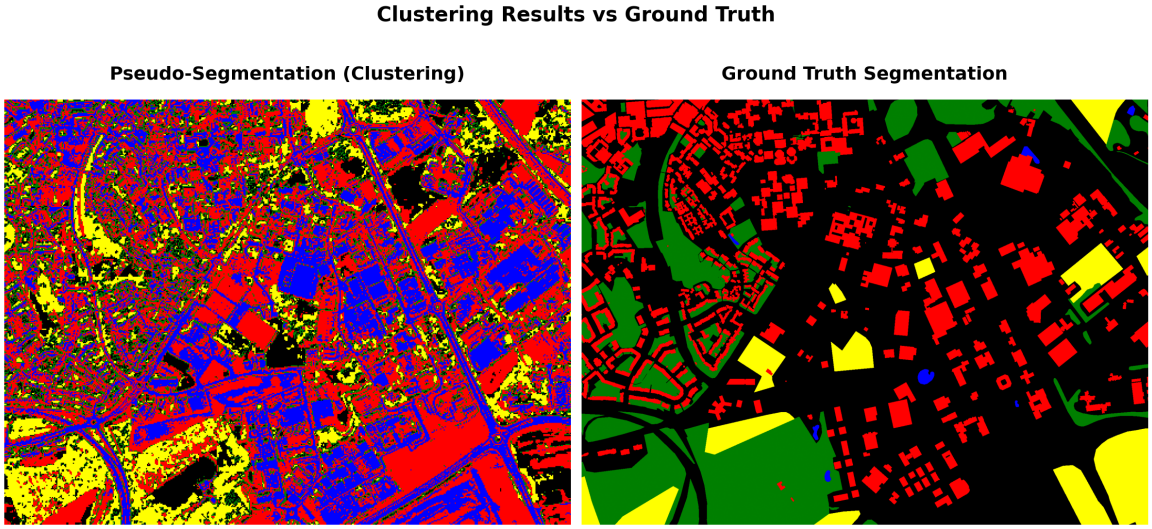


Figure 3.9: Pseudo-segmentation from colorization features (k=5): (left) clustering result, (right) ground truth.

3.2.2.3 Context Prediction (k=5)

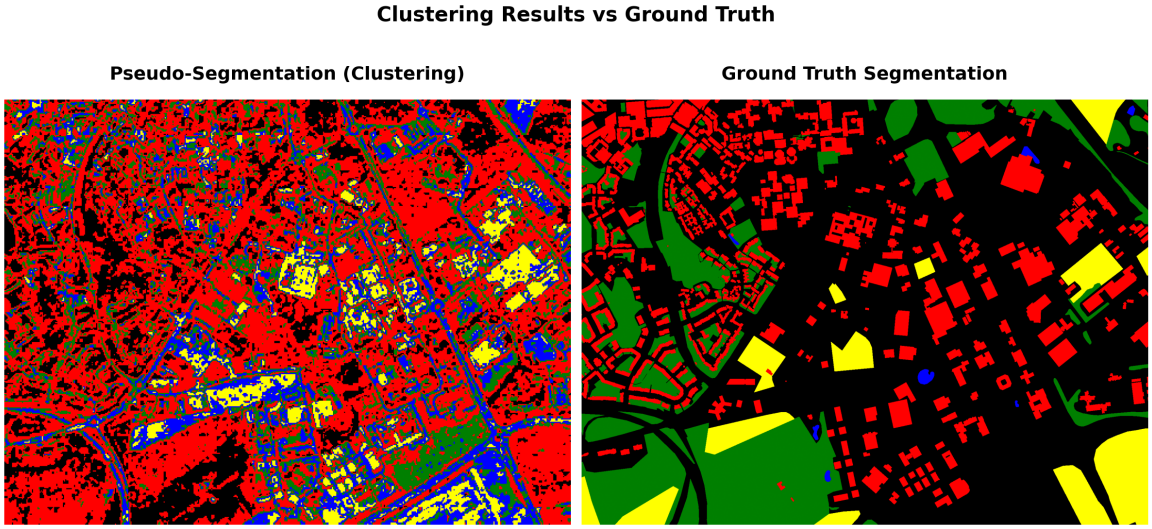


Figure 3.10: Pseudo-segmentation from context prediction features (k=5): (left) clustering result, (right) ground truth.

3.2.2.4 Denoising (k=10)

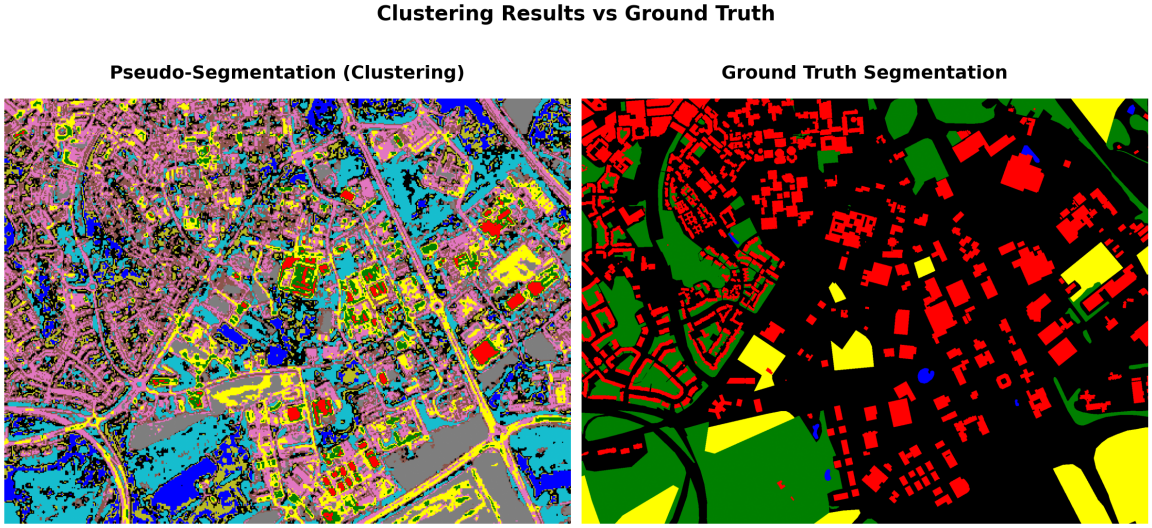


Figure 3.11: Pseudo-segmentation from denoising features (k=10): (left) clustering result, (right) ground truth.

3.2.2.5 Colorization (k=10)

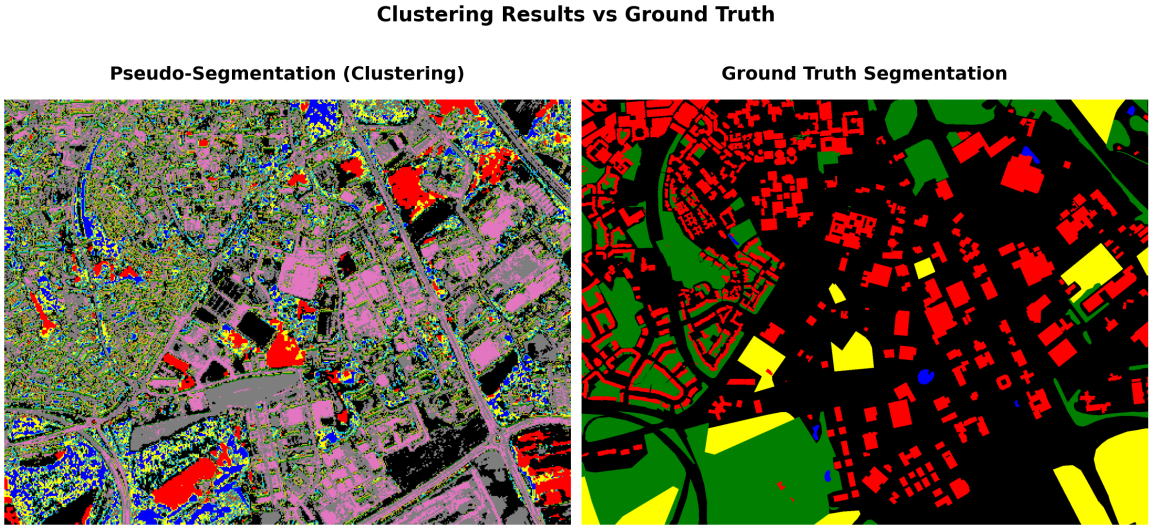


Figure 3.12: Pseudo-segmentation from colorization features (k=10): (left) clustering result, (right) ground truth.

3.2.2.6 Context Prediction (k=10)

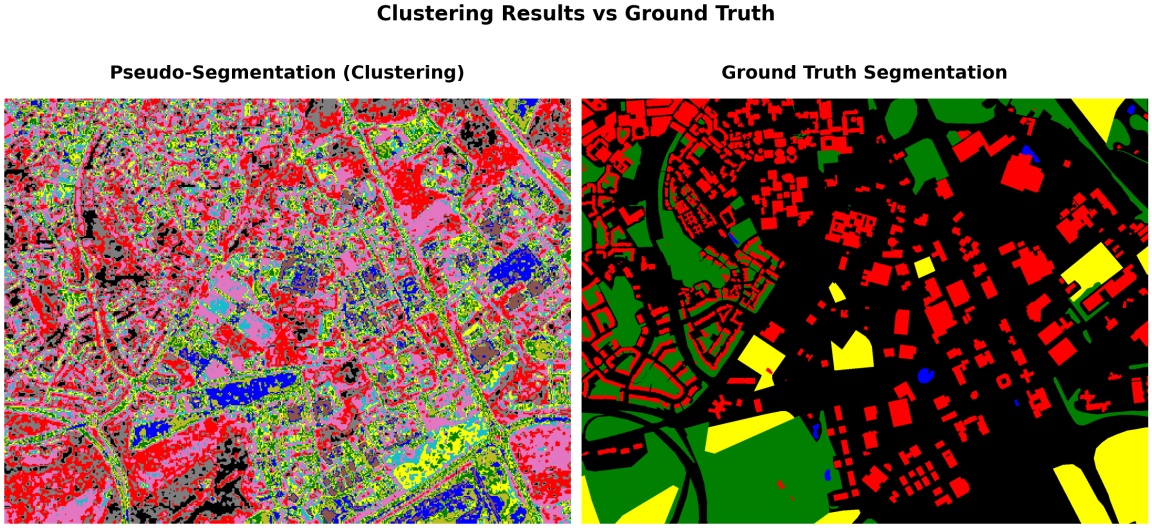


Figure 3.13: Pseudo-segmentation from context prediction features (k=10): (left) clustering result, (right) ground truth.